Beyond Accuracy: How Parameter-Efficient Fine-Tuning and Prompting Interact in Movie Question Answering

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Abstract

Long-form movie question answering (Movie-QA) requires temporal, causal, and emotional reasoning over full-length narratives, making fine-tuning language models for this task both costly and difficult to deploy at scale. We explored Parametric-efficient fine-tuning (PEFT) methods such as QLoRA, Prefix-Tuning, and ReFT to update only a small fraction of parameters and examine whether different instruction prompt styles pose a significant impact on the accuracy considering model types and question categories. Our results demonstrate that the Llama 3 8B backbone consistently outperforms alternatives in this setting. Moreover, QLoRA fine-tuning not only improves accuracy (by 8.5%) across all reasoning categories but also acts as a strong stabilizer, reducing prompt sensitivity by up to an order of magnitude. Finally, we find that prompt engineering provides a consistent marginal gain on top of fine-tuning – CoT prompting further increases QLoRA accuracy by 1.22% and Strict-Letter prompting by 1.14% – though these benefits are selective: temporal and thematic reasoning improve meaningfully while narrative retrieval categories show a slight degradation, highlighting the importance of aligning prompting strategies with task-specific reasoning requirements.

1. Introduction

Full-parameter fine-tuning large language models (LLMs) for downstream work is impractical for most research labs because it requires substantial memory and compute resources (Houlsby et al., 2019). Parameter efficient fine-tuning (PEFT) methods have emerged as effective alternatives. LoRA and QLoRA update only a small subset of parameters while keeping the majority of pretrained weights frozen. Prompt-based techniques such as Prefix-Tuning offer lightweight adaptation strategies that further improve efficiency (Petrov et al., 2023). These methods significantly reduce memory and computational requirements while maintaining competitive performance (Dettmers et al., 2023).

However, despite the growing popularity of PEFT, its effectiveness in addressing challenging reasoning scenarios, particularly "hard" questions and its interaction with prompt design remains under-explored. Base LLMs often struggle with complex temporal reasoning and exhibit high sensitivity to prompt phrasing, leading to inconsistent performance across different instruction styles (Hendrycks et al., 2021). While prior work in cinematic question answering has largely focused on multimodal inputs (e.g., video and subtitles), limited attention has been given to text-only settings and to the interplay between fine-tuning strategies and prompt formulations across diverse reasoning types. The CinePile dataset provides a unique opportunity to address these gaps by offering a structured evaluation framework across five reasoning domains and a "Hard" split. Such granularity allows for a more nuanced understanding of how different adaptation strategies perform under varying reasoning demands (Rawal et al., 2024).

In this work, we systematically investigate the impact of multiple PEFT methods across different question types and difficulty levels in a text-only MCQA setting, specifically QLoRA, Prefix Tuning, and Representation Fine-Tuning (ReFT). We further analyze how three fundamental prompt styles influence model performance, with the goal of understanding the interaction between prompting strategies and fine-tuning approaches.

2. Related Work

Traditional video QA benchmarks often focus on short clips (e.g., MSVD-QA, MSRVTT-QA). CinePile extends the horizon to full-length movies and requires long-context reasoning. It is a long video/movie QA benchmark that contains about 305k multiple-choice questions, including Temporal

Reasoning (TEMP), Character and Dialogue (CRD), Narrative and Plot Analysis (NPA), Setting and Atmosphere (STA), and Theme Exploration (TH) over full movies, additionally with a label for hard questions (Hard). CinePile's scale and fine-grained annotations make it ideal for analyzing how PEFT behaves across heterogeneous reasoning skills. In the original study, the author fine-tuned models on the training split and compared the performance with zero-shot and few-shot baselines. It turned out that the fine-tuned model outperformed with an overall accuracy improvement, especially in Narrative and Setting questions. However, there was still a significant gap between model performance and human evaluation (Zhao et al., 2021). Similarly, another research related to the long-context benchmarks, such as LongBench v2, also showed the insufficiency for deep narrative understanding along with its context and complex reasoning tasks. (Bai et al., 2025)

Another limitation is that LLMs are highly sensitive to prompt styles. Minor variations in wording or different styles can lead to differences in the model performance in structured tasks like MCQA (Zhao et al., 2021). Those inconsistencies arise from instruction styles. Chain-of-Thought (CoT) prompting has been shown to improve reasoning performance by encouraging intermediate reasoning steps (Wei et al., 2022). Still, its effectiveness varies across tasks and may introduce additional variance. Those findings show that, despite desiring outperformance for LLMs in general reasoning, it still lacks stability in tracking temporal dependencies and entity interactions across long-form inputs. In this project, our research introduces an adaptive hybrid PEFT framework tailored for long-form MCQA. We focus on the text-only subset of CinePile, using subtitles, scene descriptions, and question-answer text as input. This design allows us to isolate the effects of parameter-efficient fine-tuning on language understanding and reasoning, while keeping computational costs manageable and avoiding confounding factors from multimodal modeling. To further evaluate model robustness, we design three prompting styles: Base, Strict-Letter, and Chain-of-Thought (CoT). By comparing performance across these styles and their paraphrases, we quantify how variations in instruction affect reasoning consistency and output stability.

3. Experiments and Results

3.1 Base LLM Selection

3.1.1 Experiment Setup

To establish a robust baseline for video-grounded multiple-choice question answering (MCQA), we must evaluate architectures that span different computational paradigms. The text-only CinePile task requires models to process lengthy plot synopses and execute complex reasoning over narrative, temporal, and thematic elements. To ensure our subsequent fine-tuning experiments are built on the optimal foundation, we benchmark three distinct pre-trained architectures: DeBERTa-v3-base, Mistral-7B-Instruct-v0.3, and Llama-3.1-8B-Instruct. We evaluated the three models across the full 4,941 example CinePile test set, recording accuracy across cognitive categories, the Hard split, and total parameter counts to contextualize efficiency.

Table 1. Baseline MCQA Accuracy Performance on the CinePile Test Set in Percentage (%)

Model	Params	Overall	TEMP	CRD	NPA	STA	TH	Hard
DeBERTa-v3-base	184M	21.7	20.8	21.9	20.3	22.8	17.9	20.6
Mistral-7B-Inst	7B	55.2	34.9	57.4	55.7	61.6	52.6	32.7
Llama-3.1-8B-Inst	8B	**60.0**	**40.3**	**62.3**	**56.4**	**67.8**	**52.3**	**38.8**

3.1.2 Result and Discussion

The results (See Table 1) show a clear performance gap between the encoder-only model and the two generative models. DeBERTa-v3-base struggled uniformly, achieving an overall accuracy of 21.7%—barely above the 20% random-guessing baseline for 5-way QA. This suggests that the multi-hop narrative reasoning required by CinePile fundamentally exceeds the capacity of standard bidirectional encoding without autoregressive reasoning chains.

In contrast, the decoder-only models demonstrated strong zero-shot capabilities. Llama-3.1-8B consistently outperformed Mistral-7B across all dimensions, achieving 60.0% overall accuracy (+4.8 pp over Mistral). This advantage was particularly pronounced in the Setting and Atmosphere (STA) and Character and Dialogue (CRD) categories, where

Llama scored 67.8% (+6.2 pp) and 62.3% (+4.9 pp), respectively.
Crucially, both generative models exhibited severe vulnerabilities on Temporal Reasoning (TEMP) and the "Hard" split. Even Llama-3.1 scored only 40.3% on TEMP and 38.8% on Hard questions, underscoring the inherent limitations of base LLMs in tracking extended temporal timelines and executing structurally complex deductions
Because Llama-3.1-8B provides the highest absolute baseline and the most robust foundation across every cognitive category, we select it as the exclusive backbone for our parameter-efficient fine-tuning (PEFT) experiments in Phase 2, aiming specifically to close its critical gaps in temporal and hard reasoning.

3.1.3 Layer-Wise Vocabulary Projection Analysis

To understand how Llama-3.1-8B solves multiple-choice questions internally, we investigate the functional roles of transformer components across layers. We employ vocabulary projection (Wiegreffe et al., 2024) to map hidden representations directly to the output space, tracking where the model forms its decision.
Consider a transformer model with L layers, where we analyze a given layer $l \in 0, \cdots, L-1$. Let t_{ans} denote the sequence index corresponding to the generation of the target answer token. We define the hidden state at layer l and position t_{ans} as $h^{attn}l$ for the Multi-Head Self-Attention (MHSA) output, and $h^{mlp}l$ for the residual stream output following the Multi-Layer Perceptron (MLP).
The pre-trained language modeling (LM) head serves as a linear projection matrix $W_{lm} \in \mathbb{R}^{|V|\times d}$, where $|V|$ is the vocabulary size and d is the hidden dimension size. To evaluate the predictive confidence of intermediate layers, we project these representations directly into the vocabulary space:
For the MHSA output: $logits_{l^{attn}} = h_{l^{attn}} W_{lm}{}^{T} \in \mathbb{R}^{|V|}$
For the post-MLP residual stream
$logits_{l^{mlp}} = h_{l^{mlp}} W_{lm}{}^{T} \in \mathbb{R}^{|V|}$

Let v^{*} denote the specific vocabulary index (token ID) of the correct multiple-choice option (e.g., " A" or " B"). We isolate the corresponding scalar value for this target token:

Raw Logits:
$l_{l^{attn}} = logits_{l^{attn}}[v^{*}], l_{l^{mlp}} = logits_{l^{mlp}}[v^{*}]$
Softmax Probabilities:
$p_{l^{attn}} = \frac{exp(l_{l^{attn}})}{\sum_{v\in V} exp(logits_{l^{attn}}[v])}, p_{l^{mlp}} = \frac{exp(l_{l^{mlp}})}{\sum_{v\in V} exp(logits_{l^{mlp}}[v])}$
Following this methodology, we plot two layer-wise metrics:
Probability Trajectory: The absolute softmax probabilities $(p^{attn}l,\ p^{mlp}l)$ across layers.
Raw Logit Trajectory: The unnormalized scores $(l^{attn}l,\ l^{mlp}l)$ across layers.

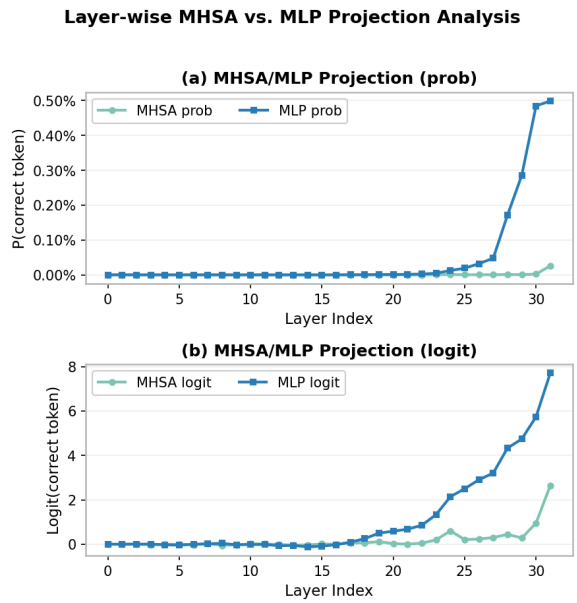

Figure 1:Layer-wise MHSA and MLP Projection Analysis.

Our projection analysis reveals a stark architectural division of labor during answer selection:
Latent Reasoning (Layers 0–17): Both MHSA and MLP projections yield near-zero logits for the target token. These layers focus entirely on aggregating scene and question semantics without manifesting explicit vocabulary preferences.
Late-Stage MLP Dominance (Layers 18–31): The post-MLP residual stream begins a monotonic climb at layer 18, accelerating sharply after layer 24 to drive the correct token's probability. In contrast, MHSA contributions remain negligible until a minor formatting spike at the absolute final layer.
Consequently, the translation of abstract reasoning into a discrete multiple-choice answer is localized

almost exclusively within the MLP matrices of the final network quartile.

3.2 Fine-Tuning Methodology

3.2.1 QLoRA: Weight-Space Modification

Low-Rank Adaptation (LoRA) updates model behavior by learning low-rank decomposition matrices W=BA injected into the linear layers of the transformer while keeping the original pre-trained weights (W) frozen. We utilize QLoRA, which further quantizes the base weights to 4-bit precision (NF4), drastically reducing memory overhead without sacrificing performance. QLoRA modifies the fundamental attention and feed-forward pathways. We expect this method to excel at shifting the model's underlying reasoning mechanisms, which is crucial for tackling structurally complex "Hard" questions and multi-step temporal dependencies.

The parameter tuning in this study is guided by recent research on the role of rank selection in low-rank adaptation. Prior work has shown that very low ranks (e.g.γ=8) are often sufficient to capture stylistic patterns but may be inadequate for injecting domain-specific knowledge, particularly in knowledge-intensive tasks such as those involving specialized legal or medical terminology (Rathore et al., 2024). This limitation arises because higher-rank adapters provide greater representational capacity to model the discrepancy between the pretrained base model and the target domain (Hu et al., 2022). However, increasing the rank introduces additional challenges. The conventional LoRA scaling factor (α/γ) can lead to unstable or diminished updates if not properly configured, necessitating careful tuning of both parameters (Kalajdzievski, 2023). At the same time, empirical evidence from QLoRA suggests that moderate ranks (e.g., $\gamma=\{8, 16, 32\}$) strike a favorable balance between efficiency and performance, particularly for instruction-following and conversational tasks (Dettmers et al., 2023). Given that multiple-choice question answering requires both reasoning and domain adaptation, our experiments adopt a rank of γ=32 as an initial configuration. Following common practice, the scaling parameter is set to α=2, ensuring sufficient update magnitude while maintaining training stability.

3.2.2 Pre-fix: Virtual-Token Modification

Unlike QLoRA, Prefix Tuning does not alter the model's weights. Instead, it prepends a sequence of learnable continuous vectors, called virtual "prefix" tokens, to the key and value states across all transformer layers.

Prefix Tuning acts as a deep, soft-prompting mechanism that steers generative behavior. Because MCQA tasks often suffer when models fail to adhere to specific output constraints (e.g., generating conversational text instead of a single letter A-E), we expect Prefix Tuning to enforce strict structural adherence and stylistic formatting at a fraction of the parameter cost of QLoRA (often requiring tuning <0.1% of total parameters).

In a previous study, we noticed that a virtual token between 10 and 50 can provide a good trade-off between efficiency and accuracy. (Li & Liang, 2021) When applying Prefix-tuning, we chose virtual token configurations to be 20 and 50 in our experiment. In addition, we enabled prefix projection, which maps the prefix embeddings through a small multi-layer perceptron to enhance the model’s adaptation to complex tasks. Additionally, in the testing experiments stage, we observed gradient exposure with the learning rate of 1e-3, which led to divergence. To address this issue, the learning rate was reduced to 5e-4, resulting in stable training dynamics and consistent convergence.

3.2.3. ReFT: Representation-Space Intervention

Representation Fine-Tuning (ReFT) (Wu et al., 2024) introduces a novel paradigm by abandoning weight and input modifications altogether. Instead, it learns low-rank linear interventions that directly edit the hidden representations (residual stream) during the forward pass at specific, targeted layers.

As established by recent interpretability literature (Wu et al., 2024), the hidden representations of pretrained LLMs already encode rich semantic and causal information. In the context of video-grounded QA, base models like Llama-3 often possess the necessary factual knowledge to answer questions but fail to properly aggregate narrative context or format their outputs into the required A–E structure. Rather than rewriting the model’s fundamental attention weights, ReFT acts as a highly targeted, inference-time intervention. By applying low-rank

edits to a small fraction of hidden representations in the middle-to-late layers, ReFT effectively steers the model's existing causal pathways toward accurate temporal and thematic reasoning. This allows us to adapt the model to complex downstream tasks with 10x to 50x fewer parameters than weight-based PEFTs.

We optimized our ReFT through a two-stage ablation. Experiment 1 applied interventions at layers 15, 24, and 28 with a rank of 16 and a learning rate of 0.002. While this improved Temporal Reasoning from the 40.3% baseline to 42.4%, overall accuracy dropped to 55.7%. This suggests that aggressive edits at earlier layers disrupt foundational linguistic processing, aligning with representation engineering literature (Zou et al., 2023; Meng et al., 2022), demonstrating that early layers extract surface syntax while middle-to-late layers govern semantic reasoning.

To prevent representational collapse, in Experiment 2, we shifted the interventions deeper to layers 20, 24, and 28, halved the rank to 8, and reduced the learning rate to 0.0002. This targeted approach successfully lifted Temporal accuracy further to 43.5% and improved performance on the Hard split to 40.1%, confirming that highly localized, deep-layer interventions are better suited for complex narrative reasoning.

3.2.4 Result and Discussion

As illustrated in Table 2, weight-space modification via QLoRA decisively outperformed Prefix Tuning and ReFT across all cognitive categories, elevating the 60.0% baseline to 68.7%. Crucially, QLoRA drove massive improvements on the Hard split by 11.0 percentage points and Temporal Reasoning (TEMP) by 15.7 points.

The underperformance of Prefix Tuning (61.8%) and ReFT (55.8%) can be attributed to their architectural constraints on a task requiring deep narrative synthesis. Prefix Tuning relies solely on prepended continuous prompt vectors; while effective for surface-level stylistic alignment, it lacks the parameter capacity to restructure the model's internal logic for multi-hop factual recall across thousands of tokens. ReFT similarly struggled, improving isolated temporal tasks but degrading overall accuracy. Because ReFT edits representations at specific layers rather than updating attention weights, it appears overly brittle for general MCQA; shifting the hidden states at layer 20 successfully steered the model toward timeline tracking but inadvertently disrupted the broader factual aggregation required for Character and Dialogue (CRD) or Theme (TH) questions.

Table 2. Comparison of PEFT Accuracy Performance in Percentage (%)

Model	Overall	TEMP	CRD	NPA	STA	TH	Hard
Llama-3.1 -8B-Inst	60.0	40.3	62.3	56.4	67.8	52.3	38.8
QLoRA	**68.7**	**56.0**	**70.3**	**70.6**	**72.1**	**65.8**	**49.8**
Prefix	61.6	45.1	63.6	64.2	65.7	62.1	41.9
ReFT	55.8	43.5	57.8	55.5	59.5	48.4	40.1

3.3. Prompt Sensitivity Framework

Recent work shows that instruction-tuned or fine-tuned LLMs are often highly sensitive to semantically equivalent instruction rephrasings. Variation in wording can result in substantial differences in model performance even though preserving meaning (Sun et al., 2023). This raises reliability concerns about the single-prompt evaluation, suggesting that instruction robustness is integral to model quality.

Our Phase 2 experiment concludes that QLoRA was our strongest PEFT method on CinePile. Phase 3 will then investigate prompt sensitivity using the QLoRA-adapted LLaMA model as the primary subject. Specifically, this phase examines how different instruction styles affect performance and robustness by answering the following questions: (1) How robust is the model to semantically equivalent paraphrases within each instruction style? (2) How does instruction sensitivity differ between the base model and the QLoRA-tuned model? (3) How does this sensitivity vary across question types and difficulty levels?

3.3.1 Experiment Set Up

We construct three semantically equivalent paraphrases for each of the following instruction styles (See Appendix): **Base** prompt is a minimal instruction to select the correct option. **Strict-Letter** instruction is an instruction explicitly enforcing a single-letter output (to prevent parsing errors). **Chain-of-Thought (CoT)** is an instruction prompting the model to reason step-by-step before answering.

This results in a total of nine prompts. Each prompt is evaluated on the full CinePile test set (4,941 examples) using both the base LLM and the QLoRA-tuned model. Model performance is measured using accuracy, defined as the proportion of correctly answered multiple-choice questions. Prompt sensitivity is quantified as the standard deviation of accuracy across paraphrases within the same instruction style.

3.3.2 Results and Discussion

Accuracy and Category-Level Gains. As shown in Figure 2, QLoRA outperforms the Base LLM across all categories, with gains concentrated where the base model is weakest. TEMP sees the largest improvement (+14.9 pp), rising from near-chance (39.9%) to 54.8%, while other categories gain a more modest 6–9 pp. Applying CoT prompting further raises overall QLoRA accuracy from 68.7% to 69.92% (+1.22 pp), and Strict-Letter prompting achieves a comparable lift to 69.84% (+1.14 pp), confirming that structured prompt styles provide a consistent marginal gain on top of fine-tuning. On the Hard split, QLoRA improves by 11.1 pp (37.9% → 49.0%), disproportionately benefiting structurally complex questions. Despite this, a persistent 23 pp gap between Hard (49.0%) and Non-Hard (72.0%) accuracy remains, indicating that structurally difficult questions are not fully resolved by fine-tuning alone.

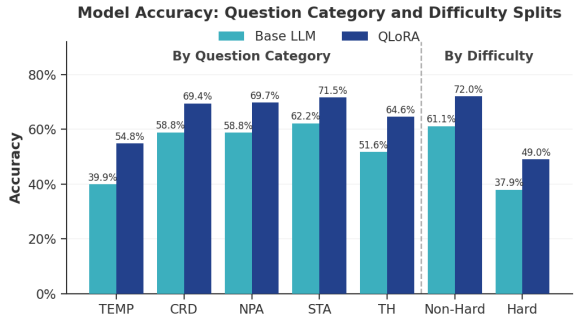

Figure 2: Model accuracy by question category and difficulty splits

Instruction Sensitivity. As shown in Figure 3, the Base LLM exhibits the highest sensitivity on TH (0.137) and TEMP (0.130) — over 1.5× the sensitivity of NPA (0.077) — revealing that abstract and temporal reasoning are most vulnerable to prompt rephrasing. QLoRA dramatically suppresses this sensitivity across all categories (TH: −63%; TEMP: −35%), demonstrating that task-specific fine-tuning yields instruction robustness as an emergent side effect of alignment, without any explicit optimization for it.

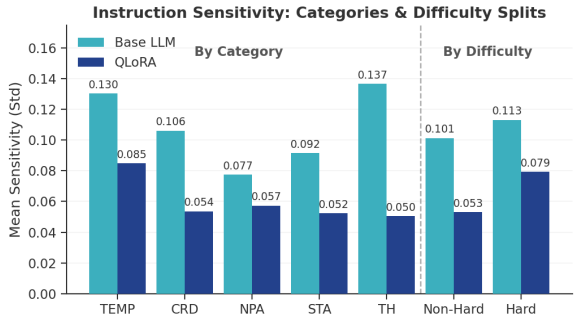

Figure 3: Prompt sensitivity by question category and difficulty splits

Selective Prompt Style Interactions. Figure 4 reveals that prompt style exerts selective rather than uniform effects. TEMP benefits most from both CoT and Strict-Letter (+2.7 pp each), and TH responds positively to CoT (+2.6 pp), confirming that structured prompting aids abstract and temporal reasoning. Conversely, NPA incurs a small CoT penalty (−1.2 pp), suggesting that over-instructing on factual plot retrieval introduces noise. These findings argue against applying a single static prompt template across all question types. For deployment, CoT or Strict-Letter is preferred for temporal and thematic questions, while a minimal Base-style prompt is safer for narrative retrieval.

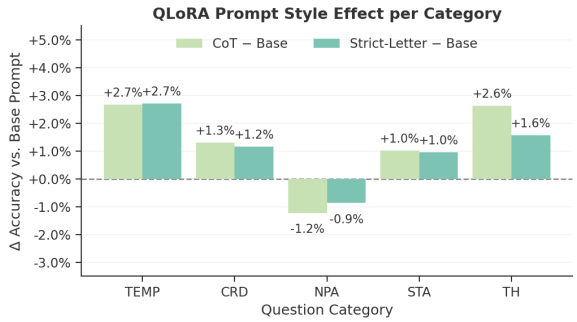

Figure 4: Accuracy lift of QLoRA's CoT and Strict-Letter prompt styles over the Base prompt, broken down by question category. TEMP and TH benefit most; NPA shows a marginal decline under both styles.

4. Conclusion

In this work, we systematically explored the text-only Multiple Choice QA problem using the challenging CinePile benchmark. After establishing Llama-3-8B as the superior base architecture in Phase 1, our Phase 2 experiments demonstrated that QLoRA fine-tuning on 30K examples yields highly competitive results, significantly lifting performance on the difficult Temporal Reasoning category and "Hard" question splits. In Phase 3, a rigorous 33 prompt sensitivity analysis revealed a crucial secondary benefit of QLoRA fine-tuning. QLoRA dramatically stabilizes the model against surface-level prompt rephrasing, making it far more robust for real-world deployment.
Beyond stability, our analysis uncovered a strict interaction between prompt style and question category. We found that Chain-of-Thought (CoT) prompting significantly boosts accuracy on logic-heavy and abstract queries—such as Temporal Reasoning and Theme Exploration—by forcing the model to explicitly map timelines and abstract concepts. Conversely, CoT actively degrades performance on factual narrative retrieval (Narrative and Plot Analysis) by introducing generative noise and hallucination risks. This insight implies that to maximize theoretical performance without additional GPU-intensive training, future systems should abandon static prompt templates. Instead, we propose the adoption of Dynamic Prompt Routing, where a lightweight classifier categorizes the cognitive demand of an incoming query and dynamically injects the statistically optimal prompt style (e.g., CoT vs. Strict-Letter) at inference time.
Overall, these results highlight that both fine-tuning strategy and prompting design play complementary roles in improving performance, with their effects varying significantly across reasoning types.

5. Next Steps

While our parameter-efficient fine-tuning (PEFT) and prompt optimization strategies yield strong improvements, critical avenues for future exploration remain. First, to close the persistent performance gap on the "Hard" split, future work must evaluate the precision-performance trade-off. Our reliance on QLoRA prioritized computational efficiency, but the 4-bit quantization of the base model can introduce minor information loss during complex reasoning passes. Evaluating unquantized LoRA on the CinePile benchmark will clarify whether the remaining error margins stem from limited adapter precision or inherent flaws in the pre-training data. Furthermore, bridging this gap on structurally difficult questions may require moving beyond single-pass generation. Exploring advanced inference-time reasoning frameworks, such as Tree of Thoughts (ToT) (Yao et al., 2023) or Multi-Agent Debate, could help models better backtrack and self-correct multi-hop logical deductions.
Finally, because this study focused strictly on the text-only setting to isolate semantic reasoning, a natural next step is to apply these PEFT intervention strategies to Vision-Language Models (VLMs)—investigating whether grounded visual embeddings can ultimately resolve the narrative ambiguities that language-only models still struggle to overcome.

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Appendix

Three paraphrases for each prompt style used in the Phase Three experiment are as follows:

1. Base:

a. You are watching a movie. Based on the scene description, answer the multiple-choice question.

b. Read the scene description and then answer the following multiple-choice question about the movie.

c. Given the scene from a movie, choose the option that best answers the question.

2. Chain of Thought:

a. You are watching a movie. Based on the scene description, think step by step about the question and options, and then answer with the letter of the correct option.

b. Read the scene description carefully, reason step by step about what is happening, and then give the letter of the best option.

c. Given the movie scene and the question, first think through the relevant details step by step, then respond with the single letter of the correct choice.

3. Strict-Letter:

a. You are watching a movie. Based on the scene description, answer the multiple-choice question. Respond with only the letter of the correct option (A, B, C, D, or E)

b. Read the scene description and select the correct option. Your answer must be a single letter from A to E.

c. Given the movie scene and question, choose the correct option and reply with only one letter (A, B, C, D, or E).

Table 3: Detailed Accuracy Breakdown

Prompt Style	Paraphrase ID	Question Category	Base LLM (Accuracy)	QLoRA (Accuracy)
Base	a	CRD	0.5948	0.6862
		NPA	0.5724	0.7041
		STA	0.6214	0.7089
		TEMP	0.407	0.5305
		TH	0.5158	0.6316
	b	CRD	0.5948	0.6862
		NPA	0.5724	0.7041
		STA	0.6214	0.7089
		TEMP	0.407	0.5305
		TH	0.5158	0.6316
	c	CRD	0.5948	0.6862
		NPA	0.5724	0.7041
		STA	0.6214	0.7089
		TEMP	0.407	0.5305
		TH	0.5158	0.6316
Chain-of-Thought	a	CRD	0.5899	0.6987
		NPA	0.6004	0.6933
		STA	0.6234	0.7167
		TEMP	0.407	0.5581
		TH	0.5105	0.6632
	b	CRD	0.5841	0.7012
		NPA	0.5853	0.6911
		STA	0.6168	0.7232
		TEMP	0.4026	0.561
		TH	0.5211	0.6579
	c	CRD	0.5827	0.6978
		NPA	0.6112	0.6911
		STA	0.6181	0.7174
		TEMP	0.391	0.5523
		TH	0.5158	0.6526
Strict-Letter	a	CRD	0.5885	0.7007
		NPA	0.6004	0.6933
		STA	0.626	0.7174
		TEMP	0.3953	0.561
		TH	0.5211	0.6526
	b	CRD	0.5861	0.6978
		NPA	0.5875	0.6998
		STA	0.6253	0.718
		TEMP	0.391	0.5596
		TH	0.5211	0.6526
	c	CRD	0.5769	0.6949
		NPA	0.5875	0.6933
		STA	0.6195	0.72
		TEMP	0.3852	0.5523
		TH	0.5105	0.6368

Contribution

Nan Wu: Contributed to the conceptualization and overall experimental design, including the three-phase evaluation framework. Established the baseline model pipeline and conducted Layer-Wise Vocabulary Projection Analysis to interpret internal model behavior. Fine-tuned the Llama-3.1-8B model using ReFT and led the analysis and visualization of prompt sensitivity results across question categories and difficulty splits.

Cassie Shi: Involved in the conceptualization of the project and developed the experimental design. Fine-tuned the LLaMa model using Prefix. Wrote the Related-Work and Experiment Part. Prepared the draft manuscript, reviewed, and edited the manuscript. Owned the Notebook: Prefix(50)-Cassie.ipynb

Yu-Sheng Lee: Involved in the conceptualization of the project and developed the experimental design. Fine-tuned Llama model using QLoRA. Ran experiments of three prompt styles and initial analysis of the result. Wrote Abstract, Introduction, PEFT and Prompty Style Experiment Methodologies, Formatted the paper.. Owned the Notebook: QLora_r32-Yusheng.ipynb, QLora_r64-Yusheng.ipynb, and prompt_styles- Yusheng.ipynb